HEART DISEASE PREDICTION REPORT

Explainable Deep Learning Diagnosis Platform (CNN-LSTM-Attention + Grad-CAM)

Patient Index	#14	**True Label (Ground Truth)**	Heart Disease
Model Prediction Probability	99.87%	**Analysis Date**	2026-06-23

Model Recommendation: HEART DISEASE PREDICTED

Patient Clinical Profile

Age	54 years	**Sex**	Female
Chest Pain Type (CP)	Asymptomatic	**Resting Blood Pressure**	120 mm Hg
Serum Cholesterol	188 mg/dl	**Fasting Blood Sugar (FBS)**	Normal (<=120 mg/dl)
Resting ECG	Normal	**Max Heart Rate (Thalach)**	113 bpm
Exercise Induced Angina	No	**ST Depression (Oldpeak)**	1.4
ST Slope	Flat	**Vessels via Fluoroscopy (CA)**	1
Thalassemia (Thal)	Reversible Defect	**Predicted Diagnosis Probability**	**99.87%**

Primary Diagnostic Drivers (AI Focus)

1. CP

Patient Value: 4.0 | **Ideal Range:** 1, 2, or 3 (Angina/Non-anginal pain)

Interpretation: *Asymptomatic chest pain (type 4). This silent presentation represents a high risk factor as blockages are painless.*

2. CA

Patient Value: 1.0 | **Ideal Range:** 0 vessels colored

Interpretation: *1 major vessel(s) colored under fluoroscopy, indicating significant arterial obstruction.*

3. THAL

Patient Value: 7.0 | **Ideal Range:** 3 (Normal)

Interpretation: *Reversible defect thalassemia observed (indicating active myocardial ischemia under stress).*

4. RESTECG

Patient Value: 0.0 | **Ideal Range:** 0 (Normal)

Interpretation: *Normal resting ECG results.*

Secondary Clinical Abnormalities (Outside Top AI Focus)

1. THALACH

Patient Value: 113.0 | **Ideal Range:** >141 bpm (85% of max predicted 166 bpm)

Interpretation: *Measured at 113.0 bpm (only 68.1% of age-predicted max). This chronotropic incompetence suggests restricted cardiac output.*

2. OLDPEAK

Patient Value: 1.4 | **Ideal Range:** 0.0 mm (No ST deviation)

Interpretation: *Measured at 1.4 mm. Significant ST segment depression suggests active myocardial ischemia during stress.*

3. SLOPE

Patient Value: 2.0 | **Ideal Range:** 1 (Upsloping)

Interpretation: *Flat/downsloping ST segment (slope type 2), strongly correlating with coronary ischemia.*

Explainability Visualization (Attention vs. Grad-CAM)

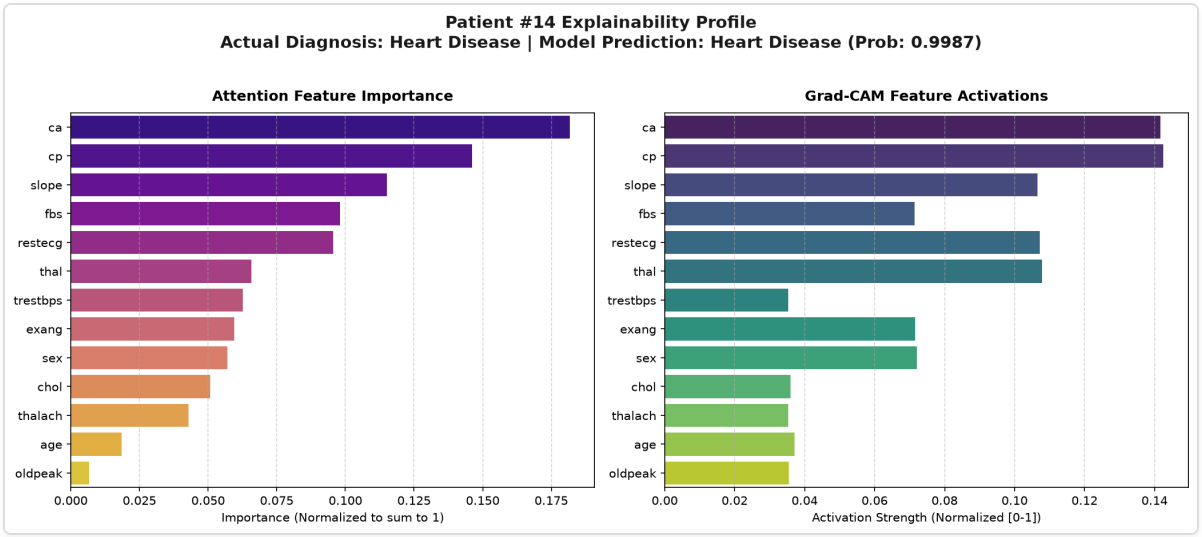


Figure 1: Dual-Perspective feature attribution showing global attention weight and patient-specific Grad-CAM contribution.

2D Gramian Angular Field (GASF) Heatmap

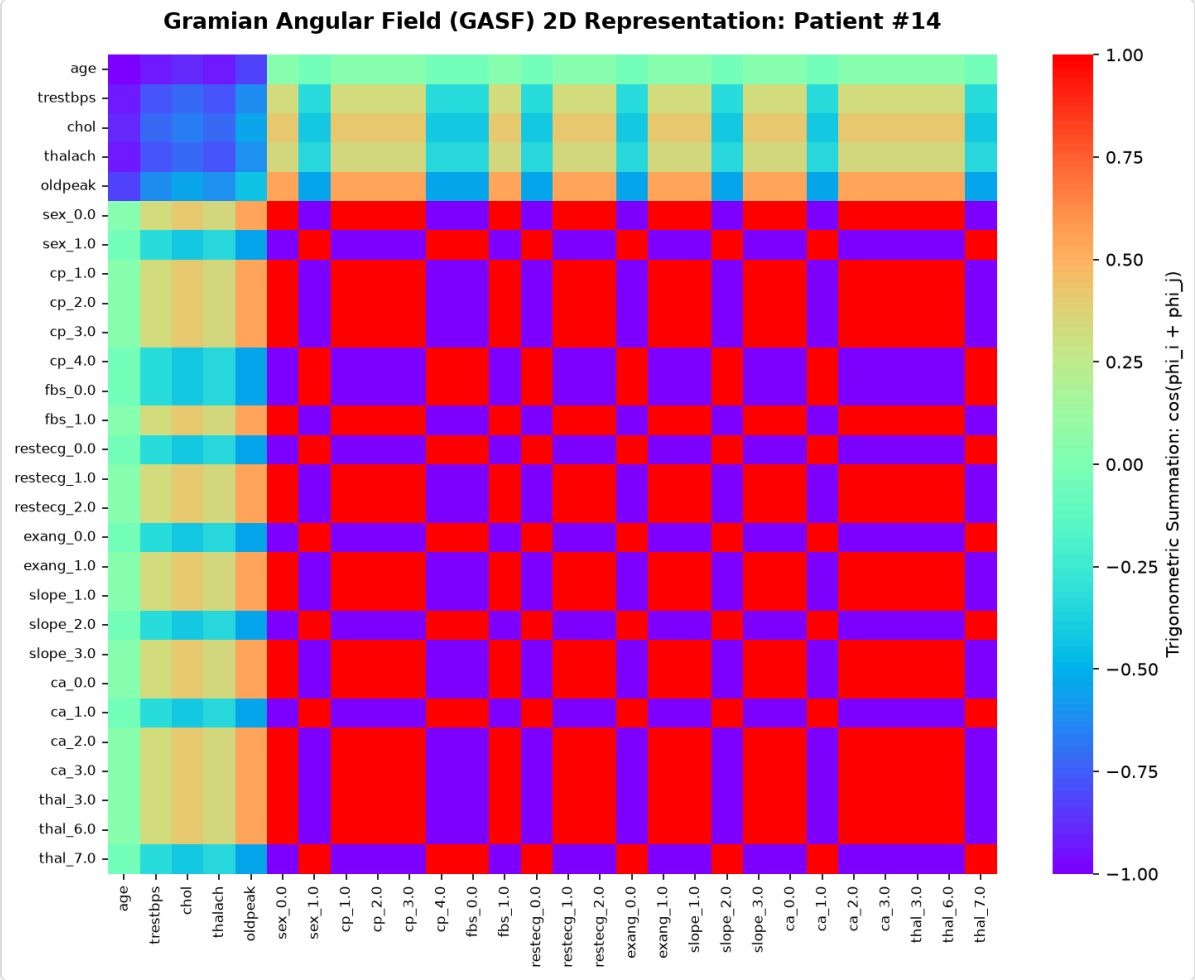


Figure 2: 2D feature-feature interaction texture mapping derived via Gramian Angular Summation Field conversion.